

RAHOOL GIR

Senior Software Engineer | Java · Microservices · Full-Stack | Fintech & Core Banking

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PROFESSIONAL SUMMARY

Software Engineer with 3 years 8 months of production experience building enterprise Core Banking and Fintech systems at TereSol Pvt. Ltd. Specialized in microservices, REST APIs, and full-stack engineering using Java, Quarkus, Spring Boot, Node.js, and Vue.js. Contributed to AL-Habib Bank's Core Banking System — building Trade Finance modules, automated transaction reconciliation, and Teller features processing thousands of daily transactions. Proficient with Docker, AWS, and SQL/NoSQL databases. Actively building AI-integrated side projects and expanding expertise in React.js, MongoDB, and modern AI development workflows.

TECHNICAL SKILLS

Backend:	Java, Spring Boot, Spring MVC, Spring Security, Quarkus, Node.js, REST API, Microservices, Maven
Frontend:	Vue.js, React.js, XState, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap
Database:	MySQL, PostgreSQL, MongoDB, SQL (Advanced)
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda), Docker, Git, GitHub, CI/CD basics
AI Tools:	Claude Code, Gemini CLI, Cursor, Antigravity, Google Gemini API, Groq API
Domain:	Core Banking, Trade Finance, Transaction Processing, Financial Workflows, SBP Regulatory Compliance

WORK EXPERIENCE

Senior Software Design Engineer

Jul 2022 – Present · 3 Yrs 8 Mos

TereSol PVT. LTD. · Karachi, Pakistan

Project: AL-Habib Core Banking System — Enterprise Production Banking Platform

- Engineered Trade Finance (Import/Export) modules for AL-Habib Bank's CBS, handling high-volume cross-border transactions in full compliance with SBP regulatory standards
- Built Teller Module features — outward bill clearing and credit card fund transfer workflows — improving transaction reliability for daily bank operations
- Automated transaction reconciliation pipelines, eliminating manual intervention in daily financial close and significantly increasing data accuracy
- Architected and consumed RESTful APIs using Java, Quarkus, and Spring Boot in a distributed microservices architecture ensuring scalable inter-service communication
- Built Vue.js frontend modules with XState for complex state management; collaborated full-stack across Agile sprint cycles

Stack: Java · Spring Boot · Quarkus · Node.js · Vue.js · XState · REST API · Microservices · MySQL · Docker · AWS

C# Developer — Trainee

Jan 2021 – May 2021

Media Monitors · Karachi, Pakistan

- Built web application modules using C# and ASP.NET MVC; enhanced data parsing routines for media content pipelines improving report generation accuracy

Stack: C# · ASP.NET MVC · SQL Server · HTML · CSS · JavaScript

Web Design Intern

Jun 2021 – Aug 2021

Abtach PVT. LTD. · Karachi, Pakistan

- Developed responsive web interfaces using HTML5, CSS3, and JavaScript; translated wireframes into pixel-perfect, cross-device UIs

Stack: HTML5 · CSS3 · JavaScript · Responsive Design

PERSONAL PROJECTS

LendLedger | 📄 [GitHub](#) · 🌐 [Live Demo](#)

AI-powered full-stack financial ledger for tracking personal debts and informal loans — digitizing the concept of "Hisaab Kitab" with enterprise-grade architecture.

- Full-stack application built with React 19 + TypeScript on the frontend and a Java Spring Boot backend architecture, featuring real-time debt tracking, premium glassmorphism UI, dark/light mode support, and Framer Motion animations
- Integrated Google Gemini AI to generate intelligent WhatsApp payment reminders, monthly financial summaries, and per-contact risk assessments — automating follow-ups that previously required manual tracking
- Implemented interactive Recharts financial dashboards and one-click PDF statement export via jsPDF, giving users professional-grade financial reports for any contact on demand

- Designed with scalability in mind: Spring Data JPA + Hibernate ORM over PostgreSQL/MySQL backend; frontend deployed on Vercel with environment-secured API keys

Stack: [React 19](#) · [TypeScript](#) · [Tailwind CSS v4](#) · [Framer Motion](#) · [Google Gemini API](#) · [Java](#) · [Spring Boot](#) · [PostgreSQL](#) · [Vite](#) · [Vercel](#)

FakeShield — AI Fake Content Detection Platform | [GitHub](#) · [Live Demo](#)

Enterprise-grade multimodal AI misinformation detection platform — evolved from a Final Year Project LSTM/CNN system (92%+ accuracy) into a full production application.

- Built a unified multimodal AI detection pipeline using Google Gemini that analyzes text articles, live URLs (via Cheerio web scraper), video frames, audio spectrograms, and images — all within a single application
- Engineered voice clone detection using spectral analysis and prosody inconsistency checks; image deepfake detection identifies GAN-generated artifacts and anatomical impossibilities
- Designed a futuristic glassmorphism UI with mouse-following aura effects, animated tab system, and full dark/light mode — combining Framer Motion with a Node.js/Express serverless backend on Vercel
- Extended original FYP (text-only LSTM/CNN detector) into a full multimodal platform with multi-provider AI routing across Google Gemini, Groq, and Ollama for redundancy and speed

Stack: [React 19](#) · [TypeScript](#) · [Node.js](#) · [Express.js](#) · [Google Gemini AI](#) · [Groq](#) · [Python](#) · [Tailwind CSS](#) · [Framer Motion](#) · [Vite](#) · [Vercel Serverless](#)

EDUCATION

B.Sc. in Computer Science 2018 – 2022

[Sindh Madressatul Islam University \(SMIU\), Karachi](#)

Key Courses: [Data Structures & Algorithms](#) · [Object-Oriented Programming](#) · [Database Management Systems](#) · [Operating Systems](#) · [Computer Networks](#) · [Software Engineering](#) · [Machine Learning & AI](#) · [Discrete Mathematics](#) · [Linear Algebra](#) · [Web Technologies](#)

Final Year Project: Fake News Detector — Deep Learning & NLP System

- Built LSTM and CNN neural network architectures trained on 40,000+ news articles, achieving 92%+ classification accuracy in detecting fabricated content
- Engineered full NLP pipeline: TF-IDF feature extraction, tokenization, stop-word removal, and word embeddings — later evolved into the FakeShield multimodal AI platform
- Developed React.js + Flask web application for real-time news verification, serving the trained ML model via REST API to end users

Stack: [Python](#) · [TensorFlow/Keras](#) · [LSTM](#) · [CNN](#) · [NLP](#) · [Flask](#) · [React.js](#) · [MySQL](#)

CERTIFICATIONS & TRAINING

Java & Data Structures and Algorithms (DSA) · [Apna College](#) Nov 2024 – Feb 2025

Topics Covered: [Java Fundamentals](#) · [OOP](#) · [Collections Framework](#) · [Exception Handling](#) · [Multithreading](#) · [Data Structures](#) (LinkedList, Stack, Queue, Trees, Graphs) · [Algorithms](#) (Sorting, Binary Search, Recursion, Dynamic Programming, BFS/DFS)

Problem Solving (Intermediate) · [HackerRank](#) Jan 2024

SQL (Advanced) · [HackerRank](#) Apr 2024

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